Week 2: Decision Trees and Random Forests

Use the same housing dataset from Week 1.

1. Theory to Research

Understand the following concepts before implementation:

- What is a Decision Tree?
- How does a Decision Tree decide where to split?
- Regression trees vs classification trees.
- Gini impurity and entropy.
- Mean Squared Error for regression trees.
- Tree depth.
- Leaf nodes and minimum samples per leaf.
- Overfitting in Decision Trees.
- What is a Random Forest?
- Why combining many trees can perform better than one tree.
- Bootstrap sampling.
- Random feature selection.
- Feature importance.

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2. Data Preparation

Start again from the original corrupted dataset.

- Identify and handle missing values.
- Fix inconsistent categories and typos.
- Remove duplicate rows.
- Correct incorrect data types.
- Encode categorical variables appropriately.
- Separate features XX and target yy.
- Create training and testing sets.

Do not use `house_id` as a feature.

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3. Decision Tree Regression

Target:

`sale_price_eur`

Train a `DecisionTreeRegressor`.

Evaluate it using:

- MAE
- RMSE
- R2R^2

Then investigate how the model changes when you modify:

- `max_depth`
- `min_samples_split`
- `min_samples_leaf`

Compare at least three different tree configurations.

Answer:

- Which configuration performs best?
- What happens when the tree becomes very deep?
- Is there evidence of overfitting?
- Which variables are most important for predicting house price?

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4. Random Forest Regression

Using the same target, train a `RandomForestRegressor`.

Experiment with:

- `n_estimators`
- `max_depth`
- `min_samples_leaf`
- `max_features`

Compare its results with:

1. Linear Regression from Week 1
2. Decision Tree Regression
3. Random Forest Regression

Create a small table containing MAE, RMSE and R2R^2 for all three models.

Explain which model performs best and why you think this happens.

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5. Decision Tree Classification

Target:

`high_price`

Predict whether a property belongs to the high-price category.

Do **not** use `sale_price_eur` as an input feature because `high_price` is directly derived from it.

Train a `DecisionTreeClassifier`.

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Experiment with different tree depths and determine whether the model starts to overfit.

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6. Random Forest Classification

Train a `RandomForestClassifier` using the same features and target.

Compare:

1. Logistic Regression from Week 1
2. Decision Tree
3. Random Forest

Use:

- Accuracy
- Precision
- Recall
- F1-score

Explain which model performs best.

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7. Model Interpretation

For the Random Forest models:

- Calculate feature importance.
- Display the top 10 most important features.
- Create a bar chart showing their importance.
- Explain whether the important features make sense for house pricing.

Final Questions

Write short answers to the following:

1. Why can a Decision Tree overfit easily?
2. How does Random Forest reduce overfitting?
3. Why does Random Forest usually perform better than a single Decision Tree?
4. What is the disadvantage of using many trees?
5. Which model worked best for house-price prediction?
6. Which model worked best for high-price classification?
7. Did the most important features match what you expected?
8. When might you prefer a simple linear/logistic regression over a Random Forest?

Use the same housing dataset from Week 1